Team 39: Same Runs, Different Strategies

DAI Course Project

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1. Main Hypothesis / Research Question

In cricket, two overs can produce the same total number of runs but through very different combinations of boundaries and non-boundary scoring. One over might derive most of its runs from fours and sixes, while another might achieve the same total through singles, doubles, triples, and extras. We aim to investigate whether these differences in scoring composition are associated with the eventual success or failure of a team in the Indian Premier League (IPL).

Our primary research question is: For a fixed number of runs scored in an over, how does the distribution of those runs between boundaries and non-boundary scoring differ between winning and losing IPL teams?

We hypothesise that the distribution of boundary counts may differ between overs bowled by winning and losing teams, even when their total runs are identical. One possibility is that winning teams tend to score more boundaries, reflecting their ability to produce high-impact shots. Alternatively, teams may achieve similar outcomes through more consistent accumulation of runs without relying heavily on boundaries. The data may also reveal that the relationship varies with the match situation rather than following one general pattern.

To investigate this question, we plan to use ball-by-ball IPL match data in JSON format obtained from Cricsheet. We will aggregate individual deliveries into overs and record the total runs, number of fours and sixes, boundary runs, non-boundary runs, and the eventual match outcome for each batting team. We will compare the distributions of boundary counts among overs with the same total runs, separately examining winning and losing teams. We also plan to explore whether the patterns differ between the first and second innings or across different phases of an innings.

Our analysis will use probability distributions, conditional probabilities, expected values, variance, visualisations, and suitable statistical tests to evaluate the observed differences. By controlling for the total runs scored in an over, we hope to distinguish differences in scoring composition from differences in scoring volume. The aim is to identify meaningful patterns in IPL batting rather than assume beforehand that one scoring approach is superior.

1.1 Source Papers, Blogs, or Podcasts

- **Cricksheet:** https://cricsheet.org/matches/

1.2 Programming Languages, Tools, and Software

Python, Pandas, NumPy, SciPy, Matplotlib, Seaborn, GitHub

1.3 Datasets

- https://cricsheet.org/format/json/

1.4 LLMs and AI Tools

- **Chat GPT** https://chatgpt.com/share/6acd2c84-5220-83ec-9857-81727ad7e611

2. Logistics

2.1 Updated Spreadsheet

https://docs.google.com/spreadsheets/d/15ZYSyDyjj8FJfHSKnPnMlt-1SqFu8Rc_NLoO75EIdv4/edit?usp=sharing

2.2 Proposed Work Breakdown and Plan

- **Teja- 25b0910:** Help interpret findings, and coordinate the final report and presentation.
- **Arangi - 25b0922:** Data collection from reliable sources and formatting, making the report
- **Kanikicharla - 25b0956:** Calculating probabilities,means,variances,confidence intervals
- **Siddhartha - 25b1000:** Calculating distributions, observing patterns, making plots and diagrams

2.3 Repository and Collaboration Links

- **GitHub repository:** https://github.com/tejayeswanth40-dev/Same-Runs-Different-Strategies.git